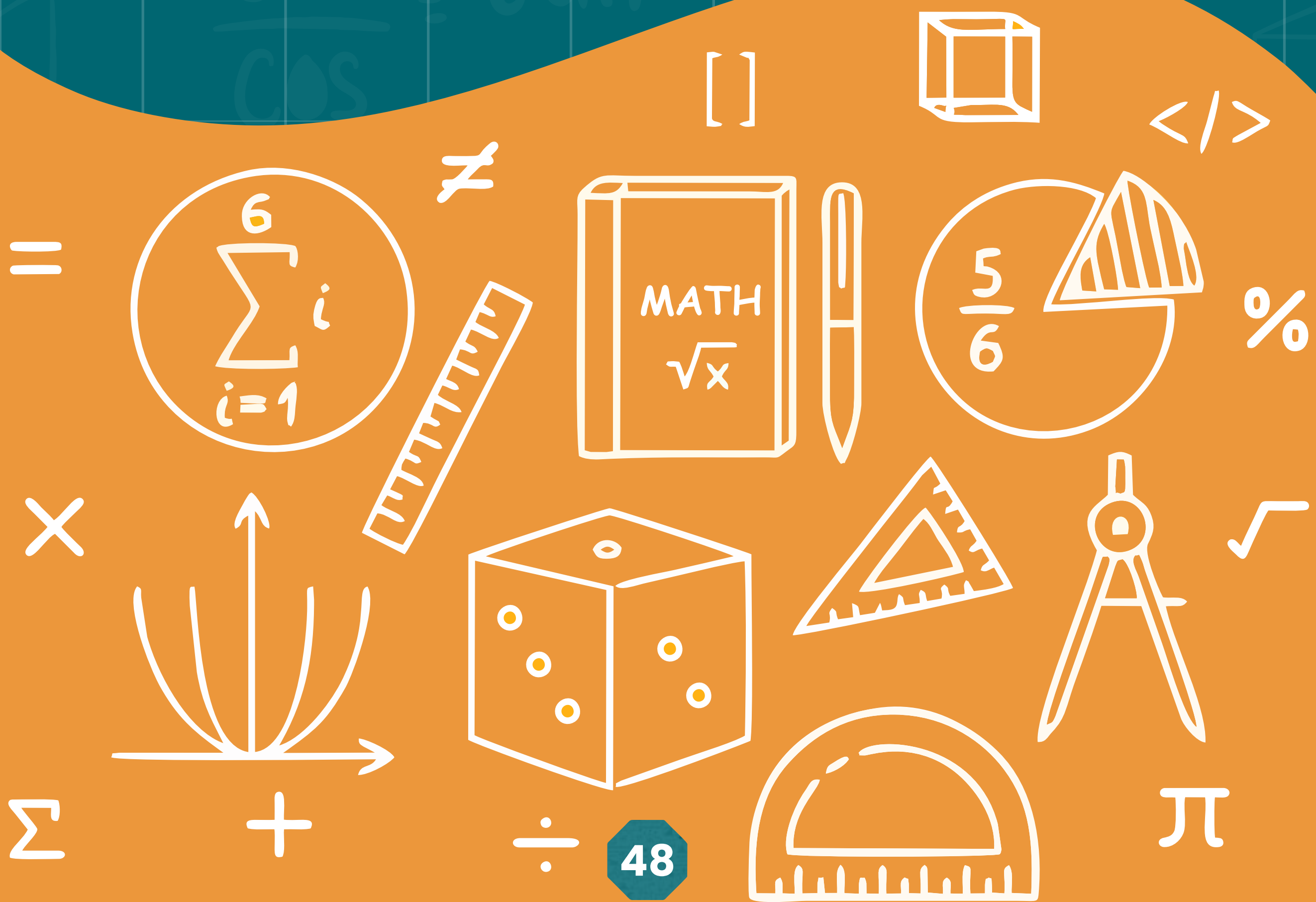
UNIT 4

FINANCIAL ARITHMETIC



FINANCIAL ARITHMETIC

Financial Arithmetic is an important branch that deals with the calculation of financial transactions and money matters. It holds fundamental significance in banking, business, and investments. In this chapter, we will learn some key concepts, formulas, and their applications.

Definitions

1. Simple Interest :

Simple Interest is a type of profit that is given on the principal amount based on a fixed rate and time period.

Formula:

$$SI = \frac{P\times R\times T}{100}$$

P: Principal Amount
R: Rate of Interest
T: Time

Example: Simple Interest Calculation

Calculate 5% simple interest for 3 years on Rs.10,000.

Solution:

Given:

P = 10,000 Rs, R = 5%, T = 3 years

$$SI = \frac{P \times R\times T}{100}$$

$$SI = \frac{P \times 10{,}000 \text{ x } 5 \text{ x } 3}{100}$$

$$SI = \text{Rs. } 1{,}500$$

Compound Interest:

Compound Interest is the profit that is given on the principal amount as well as the interest accumulated on it, after each year or specified period.

Formula:

$$CI = P\left(1 + \frac{R}{100}\right)^T - P$$

P: Principal Amount
R: Rate of Interest
T: Time

Example: Compound Interest Calculation

Calculate 5% compound interest for 2 years on Rs.10,000 ?

Solution:

Given:

P = 10,000 Rs

R = 5%

T = 2 years

$$CI = 10{,}000\left(1 + \frac{5}{100}\right)^2 - 10{,}000$$

$$= 1{,}025$$

Answer:

Compound Interest Rs. 1,025

PROFIT AND LOSS

Profit:

When the selling price (SP) is greater than the cost price (CP).

Profit = SP - CP

Loss:

When the selling price (SP) is less than the cost price (CP).

Loss = CP - SP

Example: Profit and Loss

A book was purchased for 400 Rs and sold for 450 Rs. Calculate the profit or loss.

Solution:

Given:

CP = Rs. 400

SP = Rs. 450

Formula:

Profit = SP - CP

Solution:

Profit = 450 - 400 = Rs. 50

Discount:

A discount is the amount that is subtracted from the original price of an item.

Discount = Marked Price - Selling Price

If rate of discount is given,

Selling Price = Marked Price x Discount Rate

EXERCISE 4.1

1) Calculate 8% simple interest for 4 years on Rs. 15,000.
2) Find the 6% simple interest for 2 years on Rs. 25,000.
3) Calculate 10% compound interest for 3 years on Rs. 20,000
4) Find the 5% compound interest for 2 years on Rs. 12,000
5) A bag was purchased for Rs. 800 and sold for Rs. 950 Calculate the profit.
6) A pen was purchased for Rs. 50 and sold for Rs. 40. Calculate the loss.
7) The marked price of a shirt is Rs. 1,200. If a discount of 10% is offered, calculate the selling price.
8) The price of a mobile is Rs. 30,000. A discount of 15% has been given. Calculate the discount and the selling price.

PARTNERSHIP

It is a business which two or more persons run together by sharing the profit and loss.
(i) Simple Partnership: It is the partnership, in which the partners start the business and close it together with equal or different investment capitals.
(ii) Compound Partnership: It is the partnership, in which the partners contribute different capitals for different time periods. The profit or loss is divided on the basis of the amount of investment and the period of investment of each partner.

Solve real life problems involving partnership.
Example 1. Asad and Ammar started a business with capitals of Rs 63,000 and Rs 72,000, respectively. After one year they earned a profit of Rs 45,000. Find the share of each one in the profit.
Solution: This is the case of simple partnership. As the time period of investment of both partners is the same, so the profit will be divided on the basis of ratio of the their capital share.
Ratio between the investments in rupees:

Asad		**Ammar**
63,000	:	72,000
63	:	72
7	:	8

Sum of ratios = 7 + 8 = 15
Total profit = Rs. 45,000

Asad's profit: $= \dfrac{7}{\cancel{15}_1} \times \cancel{45,000}^{3000} = \text{Rs. } 21,000$

Ammar's profit: $= \dfrac{8}{\cancel{15}_1} \times \cancel{45,000}^{3000} = \text{Rs. } 24,000$

Example 2. Amjad started a business with a capital of Rs 30,000. After 6 months Umar joined him with an investment of Rs 60,000. Usman joined them after three months by investing Rs 180,000. At the end of the year they earned a profit of Rs 490,000. Find the share of each partner in the profit.

Solution: This is the case of compound partnership in which capitals and time periods of investments of partners are different.

Amjad's effective investment = Investment capital × Time period
= 30,000 × 12 = Rs. 360,000
Umar's effective investment = 60,000 × 6 = Rs. 360,000
Usman's effective investment = 180,000 × 3 = Rs. 540,000
Ratio of effective capitals of partners in rupees:

Amjad	:	**Umar**	:	**Usman**	
360,000	:	360,000	:	540,000	
36	:	36	:	54	
2	:	2	:	3	(Sum of ratios = 2 + 2 + 3 = 7)

Amjad's Share in profit: $= \frac{2}{7} \times 490{,}000 =$ Rs. 140,000

Umar's Share in profit: $= \frac{2}{7} \times 490{,}000 =$ Rs. 140,000

Usman's Share in profit: $= \frac{3}{7} \times 490{,}000 =$ Rs. 210,000

EXERCISE 4.2

1. Marium and Arifa started a business with investments of Rs 60,000 and Rs 80,000 respectively. After one year they earned a profit of Rs 21,000. Find the share of each partner in the profit.

2. Akram started a business with Rs 70,000. After 3 months Aslam joined the business with Rs 40,000 and after 6 months Asghar also invested Rs 40,000. At the end of the year they earned a profit of Rs 28,800. Find the share of each partner in the profit.

3. Ali, Zain and Saad started a business with Rs 30,000, Rs 30,000 and Rs 24,000, respectively. After 5 months period Zain withdraws Rs 18,000 and the business is closed after 10 months. How much did each partner receive in the profit of Rs 37,500?

4. Ali and Usman invested different amounts for different periods in a business. If the total profit in a business after one year is Rs 60,000, fill in the following blanks:

S. No.	Investment of Ali (Rs)	Investment of Usman (Rs)	Period of Ali (Months)	Period of Usman (Months)	Ratio of Profits	Profit of Ali (Rs)	Profit of Usman (Rs)
(I)	50,000	25,000	12	12	2 : 1	40,000	20,000
(ii)	50,000	75,000	12	12	______	______	______
(iii)	25,000	25,000	6	12	______	______	______
(iv)	50,000	25,000	12	6	______	______	______
(v)	50,000	______	6	6	2 : 3	______	______

INCOME TAX

Explain Income Tax, Exempt Income and Taxable Income.

(a) Income Tax: Income tax is imposed by the government on the annual income of a person or company whose income exceeds a certain limit. The rules and rates for income tax are amended by the government and are announced during the presentation of annual budget.

(b) Exempt Income: It is the income which is not subject to income tax under the present income tax law. Exempted income is as follows:

(i) Agriculture Income
(ii) Gratuity or pension
(iii) Income of house or property owned by widows.

(c) Taxable Income: It is the difference of total annual income and exempted income.
Taxable Income = Total Annual Income-Exempted Income.

(d) Rebate: Government announces a certain limit of earning on which a person has not to pay income tax. This limit is called rebate. For example, if annual income of a salaried person is Rs 500,000, the rebate is Rs 400,000, then he has to pay income tax only on Rs 100,000, at the rate announced every year in the budget.

(e) Assessment Period: It starts on first July every year and ends on 30th June the following year.

Taxable Income Slabs 2016 - 2017

S. No.	Taxable Income	Rate of Tax
(I)	Where the taxable income does not exceed Rs 400,000	0%
(ii)	Where the taxable income exceeds Rs 400,000 but does not exceed Rs 500,000	2% of the amount exceeding Rs 400,000
(iii)	Where the taxable income exceeds Rs 500,000 but does not exceed Rs 750,000	Rs 2,000 + 5% of the amount exceeding Rs 500,000
(iv)	Where the taxable income exceeds Rs 750,000 but does not exceed Rs 1,400,000	Rs 14,500 + 10% of the amount exceeding Rs 750,000
(V)	Where the taxable income exceeds Rs 1,400,000 but does not exceed Rs 1,500,000	Rs 79,500 + 12.5% of the amount exceeding Rs 1,400,000
(Vi)	Where the taxable income exceeds Rs 1,500,000 but does not exceed Rs 1,800,000	Rs 92,000 + 15% of the amount exceeding Rs 1,500,000
(Vii)	Where the taxable income exceeds Rs 1,800,000 but does not exceed Rs 2,500,000	Rs 137,000 + 17.5% of the amount exceeding Rs 1,800,000
(viii)	Where the taxable income exceeds Rs 2,500,000 but does not exceed Rs 3,000,000	Rs 259,500 + 20% of the amount exceeding Rs 2,500,000
(ix)	Where the taxable income exceeds Rs 3,000,000 but does not exceed Rs 3,500,000	Rs 359,500 + 22.5% of the amount exceeding Rs 3,000,000
(x)	Where the taxable income exceeds Rs 3,500,000 but does not exceed Rs 4,000,000	Rs 472,000 + 25% of the amount exceeding Rs 3,500,000
(xi)	Where the taxable income exceeds Rs 4,000,000 but does not exceed Rs 7,000,000	Rs 597,000 + 27.5% of the amount exceeding Rs 4,000,000
(xii)	Where the taxable income exceeds Rs 7,000,000	Rs 1,422,000 + 30% of the amount exceeding Rs 7,000,000

Solve Simple Real Life Problems Related to Individual Income Tax Assesses.
Example 1. Annual income of Junaid is Rs 480,000. Calculate the amount of income tax to be paid by Junaid.
Solution: Annual income Rs 480,000This income is listed in the Taxable income slab at Sr.# 2,

where Rate of Tax is 2% of the amount exceeding Rs 400,000.
Here minimum income tax is 0 (Rs 0.0)
Income exceeding Rs 400,000 is:
Rs 480,000 - Rs 400,000 = Rs 80,000
Income tax = Rate of Tax x Income exceeding Rs 400,000

$$= \frac{2}{100} \times 80{,}000 = 1600$$

Example 2. Annual income of Ashraf is Rs 900,000. Calculate the amount of income tax, if he paid Rs 50,000 as Zakat.
Solution: Annual income - Rs 900,000, Amount Paid as Zakat Rs 50,000
Income calculated for tax 900,000 - 50,000 = Rs 850,000
This income is listed in Taxable income slab at Sr. # 4, where rate of tax is 10% of the amount exceeding Rs 750,000 which is
Rs. 850,000 - Rs 750,000 = Rs 100,000
Also minimum income tax is Rs 14,500
So, Total Income Tax Minimum Income Tax + Tax on Rs 100,000
= Rs 14500 + 10% of Rs 100,000
= 14,500 + 10,000 - Rs 24,500.
Ashraf has to pay Rs 24,500 as income tax.

Example 3. Harmain paid Rs 107,000 as income tax in a year. Find her annual income.
Solution: Income tax paid in a year = Rs 107,000
As this amount is greater than Rs 92,000 by Rs 15,000 but less a than Rs 137,000, so taxable income of Harmain falls in income slab at Sr#6, where rate of tax is 15% of the amount exceeding Rs 1,500,000.
Thus we have to find the amount exceeding Rs 1,500,000 on which Harmain paid Rs 15,000 more than Rs 92,000 as income tax.
If Rs 15 is income tax on Rs 100 then Re 1 is income tax on Rs 100/15
Rs 15,000 is the income tax on Rs 100/15 x 15,000=Rs 100,000
Hence, annual income of Harmain is: Rs 1,500,000+ Rs 100,000 = Rs 1,600,000.
Verification:
Income tax on Rs 1,600,000 = Rs 92,000 + 15% of 100,000 = 92,000 + 15,000 = Rs 107,000 (Verified)

EXERCISE 4.3

1. Monthly salary of Amjad is Rs. 75,000. Compute his annual income tax.
2. Hania earns Rs 495,000 in a year. Calculate her income tax, if she paid Rs 40,000 as Zakat.
3. A person has earned Rs 7,000,000 in a year. The tax deducted at source is Rs 120,000 and Zakat deducted is Rs 130,000. Calculate the income tax which he has to pay at the end of the financial year.
4. The monthly salary of Ismail is Rs 200,000. Find the income tax he paid, if Rs 80,000 and Rs 20,000 are paid as Zakat and wealth tax, respectively.
5. Sufyan paid Rs 497,000 as income tax in a year. Find his monthly income.
6. Bushra paid Rs 289,500 as income tax at the end of a year. If she has paid Rs 130,000 as wealth tax and Rs 120,000 as Zakat, find her annual income.
7. Complete the following table for different income categories:

S. No.	Yearly Income (Rs in 1000)	Income Tax Slab Number	Min. Income Tax (Rs)	Lower Limit (Rs)	Income Exceeding Lower Limit (Rs in 1000)	Tax Rate (%)	Total Income Tax (Rs)
(I)	578	3	2,000	500,000	78	5%	5,900
(ii)	480	______	______	______	______	______	______
(iii)	900	______	14,500	______	______	______	______
(iv)	6,000	11	______	______	2,000	27.5%	______